Ryan Katzman

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Objective

Dedicated student with a passion for engineering and mathematics, always inclined to take on difficult challenges. Seeking a Computer Engineering internship for Summer 2027.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*

Bachelor of Science in Computer Engineering, Mathematics minor, GPA 4.00 Expected Graduation, December 2029

Skills

Programming: Java, Python; C (basic), RISC-V Assembly (basic)
HDLs: SystemVerilog, VHDL (basic)
Hardware: Digital Logic, RTL design, ESP32, Breadboarding
Tools: VSCode, Quartus Prime
Coursework: Digital System Design, Circuit Analysis, Data Structures and Algorithms**,** Programming HW/SW System, Digital Design Lab
Languages: English (native), Spanish (Conversational)

Projects

***SHA-256 Cryptographic Accelerator* | Silicon Jackets** . *March 2026 – Present*

Digital Design Team

- Collaborating in a four-person team to develop a hardware implementation of the SHA-256 cryptographic hash function in SystemVerilog.
- Designing modular RTL components for the SHA-256 message schedule, compression function, and control FSM.
- Applying combinational and sequential logic design techniques to optimize datapath throughput and ensure timing closure.

***64-bit Adder* | Silicon Jackets** *January 2026 – March 2026*

- Designed a 64-bit unsigned integer adder in SystemVerilog using 32-bit operations and a multi-cycle architecture.
- Implemented datapath and finite-state machine control logic to coordinate operand reads, addition, carry propagation, and result storage.
- Executed RTL synthesis and static timing analysis in Quartus Prime to evaluate gate-level performance.

***Nourishmate* | Grand Challenges @ Georgia Tech** *August 2025 – May 2026*

- Developed a Tamagotchi-inspired physical companion device designed to encourage healthy eating habits and combat eating disorders; voted #1 class project in Grand Challenges 2025-2026.
- Integrated an ESP32 microcontroller with wireless data transfer to communicate user journal entries with an online therapist dashboard.
- Contributed to a hardware/software integration between the physical device and a Python and Java based system for aggregating and presenting user data.

Experience

Physics I Laboratory Teaching Assistant | Georgia Tech *September 2026 – Present*

- Assist students in applying introductory physics concepts and solving problems during laboratory sessions.
- Guide approximately 25 students per laboratory session through experimental procedures, problem solving, and data analysis.